

2. To Create Main Loop, Display Prompt, Read User Input, Handle Exit Conditions, Design Control Flow Diagram, Test Interactive Loop

Code:

```
GNU nano 8.7.1
#include <stdio.h>
#include <string.h>

#define MAX_INPUT 100

int main() {
    char input[MAX_INPUT];

    printf("Simple Interactive Shell\n");
    printf("Type 'exit' to quit.\n\n");

    // Main interactive loop
    while (1) {

        // Display prompt
        printf("shell> ");

        // Read user input
        if (fgets(input, sizeof(input), stdin) == NULL) {
            printf("\nInput error or EOF detected.\n");
            break;
        }

        // Remove newline character
        input[strcspn(input, "\n")] = '\0';

        // Handle empty input
        if (strlen(input) == 0) {
            continue;
        }

        // Handle exit condition
        if (strcmp(input, "exit") == 0) {
            printf("Exiting shell...\n");
            break;
        }

        // Display entered command
        printf("Command entered: %s\n", input);
    }

    printf("Shell terminated successfully.\n");

    return 0;
}
```

Result:

```
saisharan134@APL-5CD1366K5L:~/SKILL$ nano Skill2a.c
saisharan134@APL-5CD1366K5L:~/SKILL$ gcc Skill2a.c -o S2a
saisharan134@APL-5CD1366K5L:~/SKILL$ ./S2a
Simple Interactive Shell
Type 'exit' to quit.

shell> hello
Command entered: hello
shell> ls
Command entered: ls
shell> pwd
Command entered: pwd
shell> test
Command entered: test
shell> exit
Exiting shell...
Shell terminated successfully.
saisharan134@APL-5CD1366K5L:~/SKILL$
```

To Capture Keyboard Input, Handle Backspace, Process Enter Key, Manage Input Buffer, Support Multi-Character Commands, Test User Interaction

Code:

```
GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <termios.h>

#define BUFFER_SIZE 100

// Enable character-by-character input
void enableRawMode(struct termios *oldSettings) {
    struct termios newSettings;

    tcgetattr(STDIN_FILENO, oldSettings);
    newSettings = *oldSettings;

    newSettings.c_lflag &= ~(ICANON | ECHO);

    tcsetattr(STDIN_FILENO, TCSANOW, &newSettings);
}

// Restore normal terminal input
void disableRawMode(struct termios *oldSettings) {
    tcsetattr(STDIN_FILENO, TCSANOW, oldSettings);
}

int main() {

    char inputBuffer[BUFFER_SIZE];
    int position = 0;
    char ch;

    struct termios oldSettings;

    printf("Interactive Keyboard Input\n");
    printf("Type a command and press Enter.\n");
    printf("Use Backspace to delete characters.\n");
    printf("Type 'exit' to quit.\n\n");

    enableRawMode(&oldSettings);

    while (1) {

        printf("shell> ");
        fflush(stdout);

        position = 0;
        memset(inputBuffer, 0, BUFFER_SIZE);

        while (1) {

            // Capture keyboard input
            if (read(STDIN_FILENO, &ch, 1) <= 0) {
```

GNU nano 8.7.1

```
// Capture keyboard input
if (read(STDIN_FILENO, &ch, 1) <= 0) {
    break;
}

// Handle Enter key
if (ch == '\n' || ch == '\r') {
    inputBuffer[position] = '\0';
    printf("\n");

    // Exit condition
    if (strcmp(inputBuffer, "exit") == 0) {
        disableRawMode(&oldSettings);
        printf("Exiting shell...\n");
        return 0;
    }

    // Display entered command
    if (position > 0) {
        printf("Command entered: %s\n", inputBuffer);
    }

    break;
}

// Handle Backspace
else if (ch == 127 || ch == 8) {
    if (position > 0) {
        position--;
        inputBuffer[position] = '\0';

        // Erase character from terminal
        printf("\b \b");
        fflush(stdout);
    }
}

// Store normal characters
else if (position < BUFFER_SIZE - 1) {
    inputBuffer[position] = ch;
    position++;

    // Display typed character
    printf("%c", ch);
}
```

```
        printf("%c", ch);
        fflush(stdout);
    }
}

disableRawMode(&oldSettings);

return 0;
}
```

Result:

```
saisharan134@APL-5CD1366K5L:~/SKILL$ nano skill2b.c
saisharan134@APL-5CD1366K5L:~/SKILL$ gcc skill2b.c -o S2b
saisharan134@APL-5CD1366K5L:~/SKILL$ ./S2b
Interactive Keyboard Input
Type a command and press Enter.
Use Backspace to delete characters.
Type 'exit' to quit.

shell> hello
Command entered: hello
shell> ls -l
Command entered: ls -l
shell> echo hello world
Command entered: echo hello world
shell> helo
Command entered: helo
shell> exit
Exiting shell...
saisharan134@APL-5CD1366K5L:~/SKILL$
```